Assignment 4

Exercisium is a fissile material with atomic weight $A = 300$ and density $\rho = 30\,\text{gm/cc}$. Assume that Exercisium has a total cross section for interactions of $\Sigma_t = 10\,\text{barn}$ with a Studention-particle. When a Studention interacts with an Exercisium nucleus, one of three possibilities occurs:

1. The Studention is absorbed and disappears with probability $p_0 = 20\%$.
2. The Studention scatters in a random direction with probability $p_1 = 50\%$.
3. The Studention causes fission of the Exercisium nucleus, when two new Studentions are emitted in random directions, with probability $p_2 = 30\%$ (the original is absorbed).

Calculate the critical mass of a sphere of Exercisium by tracking the history of 100 Studentions, that is, start the 100 histories of the Studention located at the center of the sphere. The critical mass is the smallest mass of Exercisium in a sphere that can produce more than 10,000 Studentions.

Assume that:

- Avogadro's number is
$$N_A = 0.6 \cdot 10^{24}\,\text{atoms/mol}.$$
- Atomic density is defined as
$$\rho_A = \frac{N_A \rho}{A}.$$
- The mean free path is defined as
$$\text{mfp} = \frac{1}{\rho_A \Sigma_t}.$$
-
$$1\,\text{barn} = 10^{-24}\,\text{cm}^2.$$
- The volume of a sphere is
$$V = \frac{4}{3}\pi r^3.$$

Computation algorithm:

- Check the criticality of a sphere with radius r.
- If it is critical, decrease r and check again.
- If it is not critical, increase r and check again.
- Stop the computation iterations when
$$dr < 0.05r.$$
- Use the mean free path as an initial estimate for r.
- For every criticality check, start with 100 independent histories, each time beginning with a Studention located at the center of the sphere.
- For each history, track all Studentions in the sphere until one of the following cases occurs:

1. No Studentions remain in the sphere.
2. more than 10,000 Studentions are produced in the sphere.

- The sphere will be considered non-critical if all 100 histories end in case number 1.
- For each experiment, track the history of the original source particle, as well as all its daughters.
- Continue producing generations until case 1 or 2 is reached.

Hint: Create an array of size 3000 that stores the locations of the Studentions. This is actually a stack. When the Studention interacts and a new particle is created, add it to the stack and increase the counter by 1. When the Studention is absorbed, remove it from the stack and decrease the counter by 1.

1. Calculate the critical mass of the presented case. For every value of r that is checked, print the result: critical sphere / non-critical sphere.
2. Calculate 11 additional cases for different densities in the range between $\rho = 10\,\text{gm/cc}$ and $\rho = 100\,\text{gm/cc}$. Draw a graph and explain the results.
3. Calculate 13 additional cases for the following probability values:

P_0	P_1	$P_2 = 1 - P_0 - P_1$
0.1	0.75	
0.1	0.70	
0.1	0.65	
0.2	0.55	
0.2	0.50	
0.2	0.45	
0.3	0.35	
0.3	0.30	
0.3	0.25	
0.2	0.47	
0.2	0.49	
0.2	0.51	
0.2	0.53	

   Print P_0, P_1, P_2, and the critical mass. Try to find the parameter that best describes the graph in the form $y = f(x)$, where x is the parameter and y is the critical mass. Explain.

4. Perform additional calculations for the following probability values:

P_0	P_1	P_2	P_0	P_1	P_2
0.0	0.95	0.05	0.1	0.50	0.40
0.1	0.75	0.15	0.2	0.30	0.50
0.2	0.55	0.25	0.3	0.10	0.60
0.3	0.35	0.35	0.0	0.60	0.40
0.0	0.90	0.10	0.1	0.40	0.50
0.1	0.70	0.20	0.2	0.20	0.60
0.2	0.50	0.30	0.3	0.00	0.70
0.3	0.30	0.40	0.0	0.50	0.50
0.0	0.85	0.15	0.1	0.30	0.60
0.1	0.65	0.25	0.2	0.10	0.70
0.2	0.45	0.35	0.0	0.40	0.60
0.3	0.25	0.45	0.1	0.20	0.70
0.0	0.80	0.20	0.2	0.00	0.80
0.1	0.60	0.30	0.0	0.20	0.80
0.2	0.40	0.40	0.1	0.00	0.90
0.3	0.20	0.50	0.0	0.00	1.00
0.0	0.70	0.30			

For each row, calculate the critical radius and critical mass. Plot the critical radius and the critical mass as functions of c. Explain whether the results are determined mainly by c, or whether the separate values of P_0, P_1, and P_2 also have a visible effect.

5. Compare the results of Questions 3 and 4 with analytical approximations for bare systems.

   Compare the Monte Carlo results with the classical diffusion approximation and with the asymptotic diffusion approximation. Plot the critical radius as a function of c on the same graph for the Monte Carlo results, classical diffusion, and asymptotic diffusion.

   In addition, plot the critical mass as a function of c. It is recommended to use a logarithmic scale for the critical mass.

   Discuss the agreement between the Monte Carlo results and the analytical approximations. Explain why the classical diffusion approximation is expected to be accurate near critical multiplication and why its accuracy changes as c increases.